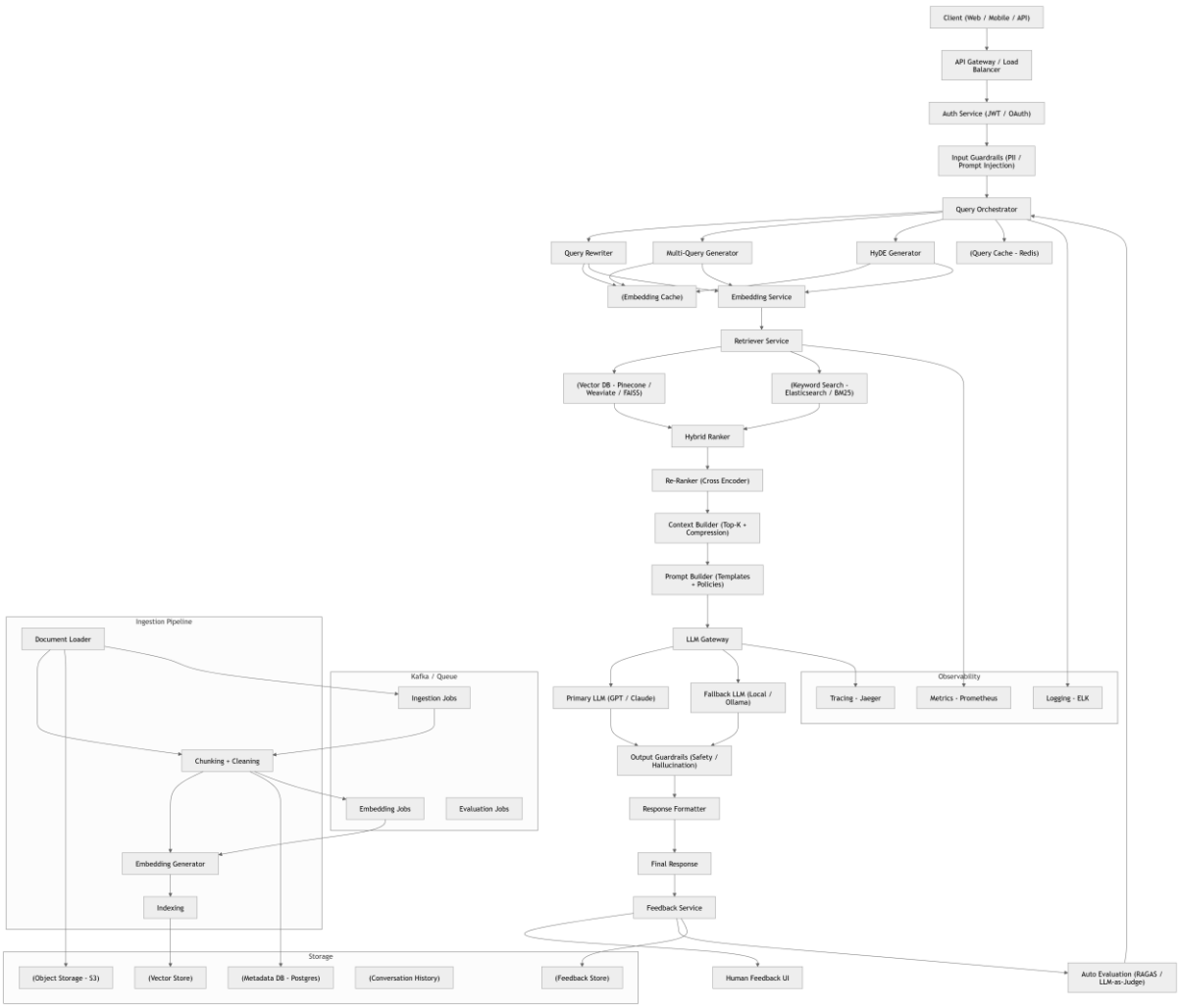
# FAANG-Level Enterprise RAG Architecture




## Suggested Tech Stack (Real-World)

| Layer | Tools |
|---|---|
| **API Gateway** | Kong / NGINX |
| **Backend** | FastAPI / Node.js |
| **Vector DB** | Pinecone / Weaviate / FAISS |
| **Search** | Elasticsearch |
| **Cache** | Redis |
| **Queue** | Kafka |
| **Storage** | S3 |

| | |
|---|---|
| **DB** | Postgres |
| **LLM** | OpenAI / Claude / Ollama |
| **Observability** | Prometheus + Grafana + ELK |

## 🔹 1. Entry Layer

- API Gateway handles routing + rate limiting
- Auth ensures secure access
- Input guardrails block malicious prompts

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## 🔹 2. Smart Query Processing

- Query rewriting
- Multi-query expansion
- HyDE (improves semantic recall)

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## 🔹 3. Hybrid Retrieval (Core of RAG)

- Vector DB → semantic search
- Elastic/BM25 → keyword match
- Combined → hybrid ranking + re-ranking

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## 🔹 4. Context Engineering (Very Important)

- Top-K selection
- Token compression
- Prompt templating

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## 🔹 5. LLM Gateway (Production Trick)

- Routes across:
  - GPT / Claude
  - Local models (fallback)
- Enables cost + reliability control

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### 🔹 6. Guardrails

- Input: injection, PII
- Output: hallucination, toxicity

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### 🔹 7. Async + Ingestion

- Kafka pipelines handle:
  - Document ingestion
  - Embeddings
  - Indexing

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### 🔹 8. Observability

- Logs, metrics, tracing

  👉 Required for debugging hallucinations + latency

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### 🔹 9. Feedback Loop (FAANG Secret Sauce)

- Human feedback
- Automated evaluation (RAGAS)
- Continuous improvement

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## ⚡ What Makes This "FAANG-Level"

✓ Microservices
✓ Scalable (Kafka + caching)
✓ Hybrid retrieval
✓ Multi-model LLM routing
✓ Full observability
✓ Continuous evaluation loop

### 1. 🔥 Microservice Separation

Each critical function is isolated:

- Query orchestration
- Retrieval
- Ranking
- Guardrails
- LLM gateway

👉 Enables **independent scaling + deployment**

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## 2. ⚡ Hybrid Retrieval (Must-have)

Instead of only vector search:

- **Vector DB** → semantic understanding
- **BM25 / Elastic** → keyword precision

👉 Combined via **hybrid ranking**

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## 3. 🧠 Smart Query Expansion

You now support:

- Query rewriting
- Multi-query generation
- HyDE

👉 Boosts retrieval recall massively

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## 4. 🛡️ Enterprise Guardrails

Split into:

- Input validation (prompt injection, PII)
- Output filtering (toxicity, hallucination checks)

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## 5. ⚡ Caching Strategy (Cost Saver)

- Redis for query caching
- Embedding reuse
- Response caching

👉 Reduces LLM cost by **30–70%**

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## 6. 📦 Async Pipelines (Kafka)

Used for:

- Document ingestion
- Embedding generation
- Evaluation jobs

👉 Ensures system doesn’t block

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## 7. 📊 Observability (Critical)

- Logs → ELK
- Metrics → Prometheus
- Tracing → Jaeger

👉 Debug latency + hallucinations

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## 8. 🔁 Continuous Evaluation Loop

- Human feedback
- Automated scoring (RAGAS)
- Feeds back into system

👉 This is how FAANG systems improve over time